



ADDENDUM — WILMINGTON MANUFACTURING CAMPUS FOR SUSTAINABLE CONSTRUCTION INNOVATION

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Why Has Nobody Brought This Manufacturing to the **East Coast** After More Than a Decade of Evidence That These Are Viable Systems?

The honest answer — and why the barriers that stopped everyone before are finally breakable in Wilmington in 2026. Plus: the Phase 2 evolution that adds 3D printed concrete and positions L.E.E.P. DE as both the training pipeline and the specialized workforce beneficiary.

THE QUESTION WORTH ASKING

If CLT, SIP, and AAC construction work so well, why is there still no regional manufacturer on the East Coast?

This is the right question — and any serious funder, city official, or private investor will ask it. The technologies are proven. Europe has operated AAC plants since the 1920s. CLT has been specified on institutional buildings globally for two decades. SIP panels have been manufactured in the US for thirty years. The evidence for performance, cost efficiency, and sustainability is not in dispute.

The honest answer is that seven structural barriers have stopped every serious attempt — and most of them are still real. Merge4Ward does not succeed by ignoring these barriers. It succeeds because the specific combination of conditions in Wilmington in 2026 changes the calculus on each one. This addendum names the barriers plainly and explains the change.

THE SEVEN BARRIERS — NAMED PLAINLY

Why every serious attempt has stalled for fifteen years.

01

Capital Requirements Are Brutal and Front-Loaded

PRIMARY KILLER

A viable CLT plant requires \$40–110 million in capital investment — before a single panel ships. The autoclave infrastructure for AAC adds \$8–20 million more. The entire capital must be committed before utilization rates are proven, before the market is developed, and before the workforce is trained. Any single funding component failing — tax credit equity, state grants, private debt — pushes the project into undercapitalization. That is the most common cause of manufacturing startup failure, and it has killed every East Coast CLT initiative that got close.

02

The Chicken-and-Egg Demand Problem

STRUCTURAL TRAP

Developers will not design projects around CLT or AAC because the regional supply chain doesn't exist. Manufacturers will not build a plant without committed order volume. Nobody moves first because the risk is entirely asymmetric — the manufacturer bears all of it. A plant that operates at 40% utilization loses money. Reaching 80% utilization requires commitments that don't exist before the plant opens. This trap has been documented in Maine, the Carolinas, and the Mid-Atlantic for over a decade. It is not theoretical.

03

The Workforce Doesn't Exist — and Training Takes Years

ADOPTION KILLER

A 2022 survey of architects and structural engineers found that lack of experience, training, and tools in the construction industry is a primary barrier to CLT adoption. For AAC, the finding is identical — contractors won't bid unfamiliar materials because they cannot price the liability. A contractor who installs an unfamiliar system and it fails owns the warranty claim. The rational response for a Mid-Atlantic contractor is to stick with wood frame. Without trained installers, manufactured panels sit in a warehouse. Merge4Ward addresses this from the founding: L.E.E.P., Inc. — a Wilmington-based 501(c)(3) affordable housing developer with an active construction pipeline — serves as the primary workforce training partner from Day 1, recruiting trainees from Wilmington's LMI neighborhoods and deploying certified graduates directly onto its own development projects. The workforce training program is not an add-on — it is the actual product without which manufacturing has no market, and L.E.E.P.'s involvement ensures that market exists before the first panel ships.

04

Timber Supply Chain Is Not Local to the East Coast

CLT-SPECIFIC

CLT requires kiln-dried, Machine Stress-Rated dimensional softwood lumber — Douglas fir, spruce, or hem-fir. The Appalachian and Mid-Atlantic forests produce hardwoods, not the softwood species that CLT presses are designed for. A Delaware CLT plant sources its lumber from the Pacific Northwest or Canada, adding back a raw material freight cost that partially offsets the panel delivery savings it was supposed to create. This supply chain reality is why every CLT facility that has opened in the US is located in the Pacific Northwest or upper Midwest — near the softwood timber base. Building one in Delaware without a plan for timber sourcing is building on an incomplete foundation.

05

06

Code Adoption Is Incomplete and Jurisdiction-Variable

REGULATORY FRICTION

IBC 2021 expanded mass timber provisions significantly — but Delaware, Maryland, Pennsylvania, and New Jersey are all at different stages of adoption. A panel manufactured in Delaware for a project in Baltimore must satisfy Baltimore's currently adopted code, which may lag IBC 2021. Each jurisdiction creates project-by-project regulatory uncertainty that slows adoption even when the technology performs. For AAC, some jurisdictions require legacy materials certification for wall systems — adding months to approval timelines. The Claymont ZBA variance was a single project milestone; scaling requires the code path to become routine, which takes years of repeated approvals building precedent.

07

Construction Industry Conservatism Is Structural, Not Incidental

CULTURE BARRIER

The building industry changes more slowly than any other capital-intensive sector. Contractors, subcontractors, inspectors, lenders, and insurers all operate in a web of established practice where "no one ever got fired for specifying wood frame." AAC has been in use globally since 1923 and still accounts for only 3% of the US walling materials market. CLT has 347 documented US projects and is still considered emerging technology by most lenders. This conservatism is rational — every party who touches an unfamiliar material takes on liability exposure their current insurance and bonding structures may not cover. Overcoming it requires not just proof of concept projects but a systematic training-and-certification infrastructure that creates a new normal. That takes a decade minimum.

Moisture Performance in the Mid-Atlantic Is Less Established

GEOGRAPHY-SPECIFIC

CLT performs exceptionally in the dry climates of the Pacific Northwest and the cold-dry climates of Scandinavia and Canada, where most of the global track record has been built. Wilmington sits in a hot-humid climate zone — summer dewpoints regularly exceed 65°F, and the Delaware River valley has persistent moisture. Moisture intrusion into CLT panel joints under imperfect sealing or ventilation can create conditions for fungal attack or wood rot. This is not a reason to abandon CLT — the hybrid design approach (CLT interior structure, SIP envelope, AAC party walls) directly addresses it by keeping CLT out of weather exposure. But it requires deliberate engineering, not assumption.

THE KATERRA CAUTIONARY TALE: The most instructive failed attempt is Katterra — a \$2 billion SoftBank-backed startup that tried to vertically integrate CLT manufacturing, design, and construction delivery across the US. It had a CLT plant in Spokane, more capital than any team proposing a Mid-Atlantic campus today, and full-stack integration that eliminated several of the barriers above. In 2021, Katterra filed for bankruptcy and shut down entirely. The CLT plant sold for pennies on the dollar. The lesson is not that the idea is wrong. The lesson is that scale, capital, and integration are not sufficient. Market timing, industry conservatism, and the sheer difficulty of changing how an entire industry builds things defeated a very well-resourced attempt. Any credible proposal for Merge4Ward must account for what Katterra got wrong — not assume it won't apply.

Why the calculus shifts in Wilmington in 2026 – and what is still not resolved.

Merge4Ward does not succeed because it is smarter than the teams that tried before. It succeeds — if it succeeds — because seven specific conditions have changed that were not true five years ago. Each condition maps directly to one of the barriers above.

X BEFORE — CAPITAL TRAP No nonprofit capital pathway existed For-profit developers raising venture capital or conventional debt faced a binary choice: commit \$40–110M before proving utilization, or don't start. No middle path.	✓ NOW — NONPROFIT STACK NMTC + CDFI + grant capital creates a staged pathway M4Di's nonprofit structure unlocks NMTC tax credit equity (\$15–40M), HUD and CDBG grants, DEDO Strategic Fund, and the Wells Fargo/Enterprise \$2M catalyst grant — a capital structure that no for-profit manufacturer can assemble. Each stage de-risks the next.
X BEFORE — CHICKEN-AND-EGG Single product, single market, binary demand CLT-only plants needed large-volume structural orders before opening. Zero orders meant zero revenue. No developer would commit before seeing the plant.	✓ NOW — DUAL-LINE DIVERSIFICATION CLT/SIP + AAC = two revenue streams, higher utilization AAC party walls serve a different market than CLT floor panels. A contractor who won't use CLT will use AAC for fire separation — it looks like concrete block. Two product lines create higher facility utilization at lower individual product volume. Still not fully solved, but materially mitigated.
X BEFORE — NO WORKFORCE INFRASTRUCTURE Panels manufactured, no trained installers to receive them Every prior attempt shipped panels to a market that didn't know how to install them. Training was assumed to follow demand. Demand never arrived because installers weren't ready.	✓ NOW — TRAINING INTEGRATED INTO THE CAMPUS L.E.E.P. + DelTech: workforce supply and demand, both solved at once The Merge4Ward workforce training program pairs Delaware Technical and Community College — curriculum and credentialing — with L.E.E.P., Inc. as the founding community partner: recruiting trainees from Wilmington's LMI neighborhoods, providing wraparound support, and deploying certified graduates onto its own development pipeline. Supply and demand-side capability develop simultaneously — not sequentially. Graduates have confirmed employment waiting before the first panel ships.
X BEFORE — NO PROOF OF CONCEPT IN THE MARKET Proposals were drawings. No Mid-Atlantic CLT building existed. Every prior East Coast CLT proposal was theoretical. No completed project. No validated cost data. No precedent-setting ZBA approval. No local lender, inspector, or code official who had seen the system in the field.	✓ NOW — CLAYMONT IS A BUILDING Panels arriving May 27. Building complete mid-July. Numbers documented. The Claymont Hybrid Townhouses are not a rendering. They are a permitted, under-construction CLT/SIP hybrid in Wilmington — with a validated GO ROM (+5.0% actual variance), a ZBA precedent, and a documented IBC compliance pathway. Every previous proposal lacked this. Merge4Ward has it.
X BEFORE — REGULATORY ENVIRONMENT UNCLEAR	✓ NOW — IBC 2021 + CLAYMONT ZBA PRECEDENT

IBC 2021 mass timber provisions did not exist. Approval was ad hoc.

Before 2021, tall mass timber construction required code variances case by case. AAC approvals were similarly project-specific. No systematic framework for either material existed.

The regulatory path exists and is documented in New Castle County.

IBC 2021 explicitly provides for mass timber construction up to 18 stories. The Claymont ZBA variance establishes the first approved CLT/SIP precedent in New Castle County. The next project starts from a known regulatory position — not from scratch.

INTELLECTUAL HONESTY — WHAT IS STILL NOT RESOLVED

The barriers that have **not changed — and must be managed, not assumed away.**

Timber supply chain: Delaware has no softwood timber base. CLT lumber will be sourced from the Pacific Northwest or Canada for the foreseeable future. The freight savings on panel delivery partially offset, but do not fully eliminate, the raw material freight premium. The campus's CLT economics are strongest when panel volume is high enough to run full truck loads on a predictable schedule — which requires sustained order volume that is not guaranteed in Year 1.

Moisture engineering in a hot-humid climate requires explicit attention at every project. The hybrid system design (CLT interior, SIP envelope, AAC party walls) is the right answer — but it must be consistently specified and inspected. One high-profile moisture failure in a Wilmington CLT building sets adoption back by years.

Industry conservatism does not change because a plant opens. It changes because a generation of contractors, inspectors, and lenders accumulates experience. The workforce training program is a direct investment in that generational shift — but it is a 10-year project, not a 3-year project. The campus should be planned and capitalized on that timeline, not on the assumption that market conversion happens quickly.

Katerra is the cautionary tale that must be internalized, not dismissed. The question for every capital and operational decision is: **how is this decision different from what Katerra did?** If the answer is not specific, the decision deserves more scrutiny.

The Fourth Technology: 3D Printed Concrete Completes the Full Building System

Phase 1 establishes the campus with CLT/SIP structural panels, AAC fire-separation panels, and modular wet wall assemblies. Phase 2 adds a 3D printed concrete fabrication bay — closing the final gap in the system and positioning Merge4Ward as the only facility in the Mid-Atlantic producing every major structural component of a high-performance affordable housing building under one roof.

WHAT 3D CONCRETE PRINTING IS — AND WHAT IT ACTUALLY DOES

A campus fabrication capability, not a fourth manufacturing line.

3D concrete printing (3DCP) uses a robotic arm or gantry system to extrude a specialized fast-setting concrete mix layer by layer, producing wall shells, foundation sections, and structural components at speeds and labor levels that conventional formwork cannot approach. A 1,500 SF home shell — walls only — can be produced in 24 hours with 3–4 operators versus the 10–12 workers conventional methods require, at material costs of \$93–\$106/SF for the printed structure.

The critical strategic distinction for Merge4Ward: 3DCP is not proposed as an on-site house printer deployed to each project site. It is a **workshop-based prefabrication capability** — the same model as CLT and AAC. The campus produces prefabricated concrete foundation wall sections, bond beam panels, ground-floor structural elements, and complex geometric components off-site, then ships them for crane-set installation. This is identical to how Citizen Robotics in Detroit operates — printing at their facility, assembling on-site. It fits the Merge4Ward model exactly.

WHAT IT ADDS TO THE SYSTEM

Closes the Concrete Components Gap

Phase 1 produces CLT floors, SIP walls, AAC party walls, and modular wet walls. It does not produce foundations, bond beams, or ground-floor concrete elements. A 3DCP bay fills that gap — making Merge4Ward a complete off-site supplier for the full building shell, not just the panels above grade.

CAPITAL ADDITION — INCREMENTAL, NOT TRANSFORMATIVE

\$800K–\$2M Added to Campus — Not \$40M

A workshop-scale robotic arm and mixing system capable of producing foundation components and prefab panels costs \$800K–\$2M installed — a manageable Phase 2 addition to an existing facility, not a new capitalization event. Equipment grants from HUD's housing technology program and DOE's advanced manufacturing portfolio are directly applicable.

WHERE 3DCP IS NOT THE ANSWER — YET

Honest Limits for Mid-Atlantic Row Houses

3DCP printed walls have lower compressive, bending, and tensile strength than conventionally cast concrete — limiting multi-story structural applications today. A 3-story attached R-3 row house is at the edge of current demonstrated capacity. CLT floors and SIP envelopes remain the right structural answer for Merge4Ward's primary project type. 3DCP handles foundations and concrete elements, not the full structural system — for now.

THE UNIVERSITY OF DELAWARE RESEARCH PARTNERSHIP: Virginia Tech, Penn State, and Ohio State all have active HUD-funded 3DCP research programs. The University of Delaware — 15 miles from Wilmington — does not. A Merge4Ward campus with a 3DCP fabrication bay producing real project components and feeding cost data into the ALDDB is a natural anchor for a U of Delaware research partnership. That partnership opens DOE Advanced Manufacturing grants and NSF construction technology programs that CLT/SIP manufacturing alone cannot access — and it makes the campus a regional knowledge center, not just a production facility.

THE BUILD SEQUENCE — WHAT OPENS WHEN

Three phases. Each funds and de-risks the next.

PHASE 1 · 2026-2027 · ACTIVE NOW CLT/SIP + AAC + Modular Wet Wall \$19M-\$49M · NMTC + Wells Fargo + DEDO Facility site identified. Dual production lines commissioned. First workforce cohort (20 trainees, L.E.E.P. pipeline supply). First panels delivered to 3 pipeline projects. ALDDB begins accumulating Mid-Atlantic CLT and AAC cost data.	PHASE 2 · 2028-2029 · 3DCP ADDITION 3D Printed Concrete Fabrication Bay \$800K-\$2M incremental · HUD + DOE grants Workshop-scale robotic concrete printer commissioned. University of Delaware research partnership formalized. L.E.E.P.-trained 3DCP operators certified and deployed. Foundation panels, bond beams, and complex concrete elements added to campus product catalog. Phase 2 workforce cohort expands to 40 trainees.	PHASE 3 · 2030+ · FULL SYSTEM Complete Building Systems Campus Operating revenue + licensing Merge4Ward produces every major structural component of a Mid-Atlantic affordable housing building under one roof: CLT floors, SIP envelope, AAC fire separation, modular MEP walls, prefab concrete foundations. 80+ graduates annually. Affiliate licensing to Baltimore, Philadelphia, D.C. Builder's Manual published nationally.
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L.E.E.P. DE: Both the Pipeline and the Beneficiary

L.E.E.P. is not simply a community partner listed on the application. It occupies two structural roles in the Merge4Ward model simultaneously — as the primary supplier of trainees into the workforce program, and as the primary beneficiary of the certified specialist workforce that program produces. This closed loop is what makes the Merge4Ward training model fundamentally different from workforce programs that train workers for jobs that don't yet exist locally.

ROLE 1 — SUPPLIER · TRAINING PIPELINE

L.E.E.P. feeds the workforce incubator

- › L.E.E.P.'s existing community relationships and workforce development networks in Wilmington's LMI neighborhoods provide the primary trainee pipeline for each Merge4Ward cohort
- › L.E.E.P. identifies, recruits, and supports candidates for CNC machinist, AAC panel fabricator, CLT assembly technician, modular wet wall installer, and 3DCP operator training tracks
- › L.E.E.P.'s 501(c)(3) structure enables it to access wraparound support funding — DDDS, DHSS, workforce development block grants — that keeps trainees in the program through completion
- › L.E.E.P.'s developer track record (Claymont Hybrid as proof) gives trainees a direct line of sight to employment on real projects — not hypothetical future jobs

ROLE 2 — BENEFICIARY · CERTIFIED SPECIALIST EMPLOYMENT

L.E.E.P. projects employ the graduates

- › L.E.E.P.'s affordable housing development pipeline — current and future — becomes the primary deployment ground for Merge4Ward-certified specialists in CLT installation, AAC party wall systems, modular MEP, and 3DCP concrete components
- › Every new L.E.E.P. development built with Merge4Ward-produced components employs Merge4Ward-trained workers — keeping both manufacturing revenue and workforce wages in Wilmington
- › L.E.E.P. gains a sustained competitive advantage as a developer: lower construction costs from local panel supply, faster schedules from trained crews, and CDFI-credible cost documentation from M4Di's predevelopment intelligence platform
- › In Phase 2, L.E.E.P. becomes the first Mid-Atlantic affordable housing developer with in-house access to 3DCP-certified operators — a workforce capability no competing developer in the region can match for years

WHY THIS MATTERS FOR THE MAYOR'S OFFICE: The L.E.E.P. dual-role model answers the question that every workforce development skeptic asks: "Who actually hires the graduates?" The answer here is specific, documented, and local — L.E.E.P., Inc. hires them, on projects already in the pipeline, building affordable housing in Wilmington. The training program has a guaranteed employment exit. The manufacturing campus has a guaranteed installer base. The city has a workforce development story it can point to with names, project addresses, and job titles attached.

The barriers are real. The window is real. The system is complete. And the window won't stay open.

No one has built a CLT/SIP/AAC manufacturing campus on the East Coast because the barriers described above are genuinely hard — capital-intensive, market-timing dependent, and culture-change slow. SmartLam, the largest US CLT producer, is right now building a second facility targeting the Eastern seaboard from Maine. The window in which Wilmington can be the regional anchor rather than a downstream customer of a Maine plant is measured in months, not years.

Phase 2 extends that advantage further: the addition of a 3D printed concrete fabrication bay — incremental capital, outsized strategic impact — makes Merge4Ward the only campus in the Mid-Atlantic producing the complete affordable housing building system under one roof. And the L.E.E.P., Inc. dual-role model — as both the trainee pipeline and the certified specialist employer — closes the loop that every prior workforce development program left open. The workers are trained. The projects are waiting. The manufacturer is Wilmington-based.

The case for city support is not that Merge4Ward is risk-free. It is that **the Mayor's Office takes zero financial risk** — a letter of support, a DEDO introduction, a Catalyst Fund conversation for WHA. Those four actions cost the city nothing. They unlock a capital stack that requires no city appropriation. And they position Wilmington — not Camden, not Trenton, not Baltimore — as the city whose Mayor had the foresight to move when the window opened.

The barriers are real. The proof of concept is built. The nonprofit capital structure is assembled. The workforce training partner — L.E.E.P., Inc. — is identified, engaged, and positioned as both supplier and beneficiary. The regulatory precedent exists. The Phase 2 evolution to 3DCP is scoped, costed, and grant-eligible from Day 1. **What is missing is the civic endorsement that signals to every funder, every investor, and every developer in the Mid-Atlantic that Wilmington is serious.** That is what the four asks in the main brief request — and why they are worth saying yes to.